



Clinical Research Paper

Thoracoscopic intercostal nerve thermal radiofrequency ablation for postoperative pain relief following the Nuss procedure



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ABSTRACT

Background: The Nuss procedure for pectus excavatum causes significant postoperative pain due to forceful correction of the sternum and ribs. Intercostal cryoablation (ICA) has shown efficacy but is limited in Japan due to insurance restrictions. This study evaluates the efficacy and safety of thoracoscopic intercostal nerve thermal radiofrequency ablation (INTRA) as an alternative postoperative analgesic approach.

Methods: A retrospective review was conducted on 170 consecutive patients who underwent the Nuss procedure with INTRA at Nishinomiya Watanabe Cardiovascular Center between May 2022 and December 2024. INTRA was performed using a radiofrequency generator and pole needles at 70 °C for 60 s. Under thoracoscopic guidance, the needles penetrated the chest wall, and ablation targeted the main intercostal nerve trunk and lateral branches. Postoperative pain control, complications, and outcomes were analyzed.

Results: The median age was 17 years (range: 12–66); 139 were male and 31 female. The mean operative time, including INTRA, was 104.8 ± 33.3 min. No procedure-related complications occurred except for minor intercostal hematomas, which resolved spontaneously. No chest tube was required for postoperative pleural effusion. Neither opioids nor epidural analgesia were needed, and 75.9 % of patients ambulated independently by postoperative day one. Among 138 patients with >6 months of follow-up, 8 % reported mild paresthesia, while the remainder had normal skin sensation. No neuropathic pain was observed.

Conclusion: INTRA is a safe and effective method for postoperative pain control in Nuss procedure patients, enabling early mobilization without opioids or epidural anesthesia. Comparative studies with ICA are warranted.

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1. Introduction

Minimally invasive surgical treatment for pectus excavatum, known as the Nuss procedure, was introduced in 1998 [1] and has since undergone significant advancements [2,3]. The depressed sternum is elevated using metal bars without requiring cartilage removal, leading to improved rib cage formation. However, the

procedure induces severe postoperative pain due to the forceful correction of the sternum and ribs. Effective postoperative analgesia is therefore essential.

The most commonly used approach for pain management includes a combination of opioid administration and thoracic epidural anesthesia [4]. Although epidural catheters provide effective analgesia during placement, they are typically removed within a few days postoperatively, after which persistent chest pain remains a significant issue.

Intercostal cryoablation (ICA) has recently been introduced as an analgesic modality for the Nuss procedure [5,6]. Previous studies have demonstrated that ICA reduces postoperative opioid consumption and shortens hospital stays, highlighting its efficacy

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as an analgesic technique [7–9]. However, in Japan, ICA is not covered by government health insurance and therefore cannot be used.

Thermal radiofrequency ablation (TRA) is an established technique for managing various chronic pain conditions by disrupting nerve transmission [10,11]. It has been widely utilized in pain clinics and has been reported to be effective for persistent intercostal neuralgia following chest trauma [12,13]. However, to our knowledge, no studies have investigated the intraoperative application of TRA as a postoperative analgesic measure. Consequently, there is limited understanding of its procedural aspects, effectiveness in acute postoperative pain relief, and associated complications.

This study aims to evaluate the postoperative analgesic efficacy of thoracoscopic intercostal nerve thermal radiofrequency ablation (INTRA) during the Nuss procedure. Additionally, we retrospectively analyze intraoperative and early postoperative complications, as well as long-term postoperative outcomes.

2. Materials and methods

INTRA was performed with informed patient consent in 170 consecutive pectus excavatum cases treated with the Nuss procedure at Nishinomiya Watanabe Cardiovascular Center from May 2022 to December 2024. A retrospective chart review was conducted to assess intraoperative complications, postoperative analgesic use and efficacy, and long-term complications in patients who had undergone surgery more than six months prior. Postoperative outpatient care was provided by one of the authors (SU), with follow-up appointments scheduled within one month of discharge, then at 3 months, 6 months, 1 year, and 2 years. At each visit, the patient was interviewed and examined by palpation regarding skin numbness or sensory abnormalities.

Since this surgical technique was unreported, with unknown efficacy and long-term outcomes, all patients were fully informed of the procedure, its potential benefits, and possible postoperative complications, and gave their consent prior to surgery. This study was approved by the Ethics Committee of the authors' institution (Approval number; 2022008).

2.1. INTRA procedure

The procedure utilized the Top Legion Generator TLG-20. A 20G, 150 mm pole needle with a 10 mm active tip was employed, with three needles used simultaneously. Differential pulmonary ventilation was not performed. Following thoracoscopic insertion via

lateral chest access with CO₂ insufflation at 5 cm H₂O, three pole needles were introduced one by one percutaneously through the mid-axillary line into the thoracic cavity under thoracoscopic guidance (Fig. 1a). After the pleura was penetrated through the intercostal muscle, the needle tip was positioned near the trunk of the intercostal nerve at 3rd, 4th and 5th intercostal spaces. The needle was then advanced slightly through the pleura along the inferior margin of the rib. Because the needle tip is sharp, particular attention must be paid to prevent lung injury; therefore, it is crucial to maintain the needle in a stable position throughout the procedure. The intercostal nerves are located within the shallow subpleural space, and excessive advancement of the needle into the muscle layer may cause deviation from the nerve. Moreover, contact of the needle tip with the intercostal artery or vein situated along the lower border of the rib may prevent adequate elevation of the probe temperature. Thus, deep needle insertion should be avoided (Fig. 1b).

After the first 60 s of coagulation, the needles were then repositioned to the upper margin of the ribs in the same intercostal space in order to coagulate the lateral branches. The third coagulation was performed in two parts in the sixth and one in the seventh intercostal space. After each coagulation, levobupivacaine and hydrocortisone were injected.

Initially, the active tip temperature was set at 80 °C for the first 10 cases but was subsequently reduced to 70 °C. Ablation was performed bilaterally from the third to seventh intercostal spaces for 60 s. In the first 26 cases, ablation was limited to the main intercostal nerve trunk at the lower border of the ribs. However, due to postoperative pain complaints, modifications were made in the subsequent 144 cases, wherein ablation targeted both the inferior and superior margins of the ribs to encompass the lateral intercostal nerve branches [14].

To prevent neuropathic pain and enhance early postoperative analgesia, 2 ml of levobupivacaine and 10 mg of hydrocortisone were injected into each treated intercostal space following ablation [15].

3. Results

The median age of the 170 patients was 17 years (range: 12–66 years), with 139 males and 31 females. Of the 170 cases, 98 (57.6 %) were under 20 years old and 34 (20 %) were in their 20s. Therefore, approximately 80 % of cases were young people. Four cases were in their 50s and two in their 60s (Table 1). The mean Haller index [16] was 5.17 ± 2.17 , and the mean correction index [17] was 32.6 ± 14.4 . The average body weight was 53.8 ± 10.5 and average

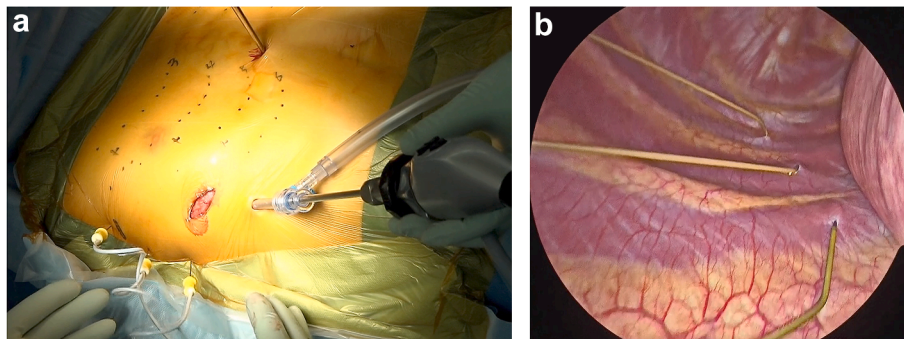
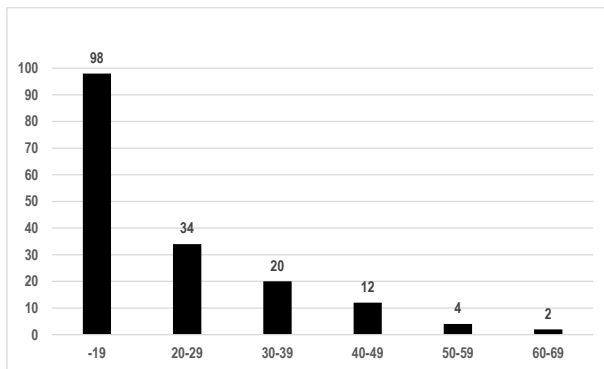


Fig. 1. a. Using the thoracoscope, three pole needles were inserted through the fourth, fifth and sixth intercostal spaces into the right chest cavity along the mid-axillary line. b. This figure shows the tips of three needles punctured the third, fourth and fifth intercostal muscles near the rib angle at the lower rib lines, with the aim of reaching the trunk of the intercostal nerves. After 60 s of ablation, levobupivacaine and hydrocortisone, were injected. The tips of the needles were then repositioned to the upper margin of the ribs in the same intercostal space in order to coagulate the lateral branches.

Table 1
Number of cases by age group.



MBI was 18.9 ± 2.7 . The mean operative time, including INTRA, was 104.8 ± 33.3 min. One pectus bar was used in three cases, two bars in 92 cases, and three bars in 75 cases (Table 2).

One patient developed a postoperative hemothorax of unknown etiology, which was managed with drainage and transfusion; spontaneous hemostasis was achieved. A thoracic drain was required in only four cases due to lung adhesion dissection and in one case due to minor lung injury-induced pneumothorax. None of these complications were attributed to the pole needle.

Complications specifically related to INTRA included cases of intercostal hematoma at the puncture sites, which resolved spontaneously without further intervention. No other INTRA-related complications were observed.

The first five patients underwent simultaneous epidural analgesia, whereas the remaining 165 patients did not receive epidural anesthesia. Early postoperative analgesia was managed with continuous intravenous buprenorphine (0.01 – 0.02 mg/h) in 91 patients (53.5 %) until postoperative day one. Postoperative oral analgesia consisted of celecoxib and loxoprofen, which were required for two to three weeks in most patients. Intravenous acetaminophen was administered as needed for three days. No opioid analgesics, including fentanyl or morphine, were administered postoperatively. Diazepam was administered at 5 mg daily for 3 days.

Among the 144 patients who underwent ablation of both the main trunk and lateral branches of the intercostal nerves, 75.9 % (85 patients) were able to walk to the toilet independently on postoperative day one. Discharge was scheduled for postoperative day six and was achieved in 97 % (165/170) of patients.

Postoperative analgesic use was discontinued within one month in 75 % (117/156) of patients with available records. Among 138 patients including the initial group of 26 patients who underwent main trunk ablation with over six months of follow-up,

8 % (11 patients) exhibited mild paresthesia in a limited area of the anterior thorax. However, in all the remaining cases overall sensory perception returned to normal after three months of surgery. Six patients experienced temporary skin hypersensitivity, which resolved within three months. No cases of neuropathic pain were reported.

4. Discussion

Thermal radiofrequency ablation (TRA) is widely used in pain clinics for managing chronic pain, including trigeminal neuralgia, low back pain and hip pain [10,11]. It has also been employed in treating post-traumatic intercostal neuralgia [12,13]. However, to our best knowledge, this study represents the first report of intraoperative application of TRA for postoperative pain control.

The findings suggest that INTRA provides effective postoperative analgesia without the need for epidural anesthesia and administration of opioids. Our short- and long-term outcomes are similar to those reported by Fraser et al. [18]. Based on our experience, traditional epidural analgesia, typically maintained for three to four days postoperatively. After epidural anesthesia is discontinued, chest pain gradually improves but may persist for several months requiring oral pain medication. The pain may interfere with daily activities and exercise. In contrast, INTRA enabled effective pain management without epidural anesthesia and opioids, facilitating early ambulation.

Previous reports on postoperative pain management using cryoanalgesia have shown that, compared with conventional methods, this technique can shorten postoperative hospital stays and reduce opioid consumption [7–9]. In our series, postoperative opioid use was successfully discontinued; however, shortening the hospital stay was not originally intended. This difference is largely attributable to the characteristics of Japan's healthcare system. Healthcare systems vary among countries, and the proportion of medical costs borne by patients also differs. Under Japan's national health insurance system, patients are responsible for 10–30 % of total medical expenses. Moreover, when monthly medical expenses become high, the patient's out-of-pocket payment is capped at approximately ¥100,000–¥200,000 (about \$670–\$1330), depending on income level. Higher-income individuals pay more, whereas lower-income individuals typically pay around ¥50,000 (about \$330). Children under 18 years of age generally have minimal or no out-of-pocket expenses. Even if the hospital stay is extended by a few days, the amount the patient pays remains unchanged.

The Nuss procedure causes substantial trauma to the entire thoracic cage, and a longer recovery period is often required before patients can resume normal activities. In addition, many of our patients had a high Haller index (mean, 5.17), indicating severe chest wall deformity, which inherently increases surgical invasiveness. Therefore, discharge on postoperative day 6 reflected the patients' preference to remain hospitalized until they felt adequately recovered to return home.

Regarding safety, INTRA carries a risk of intercostal vessel injury during needle insertion. However, our experience showed only minor hematomas that resolved spontaneously. No lung injuries were observed across 170 cases, indicating the procedure's safety when performed with careful thoroscopic guidance.

The needle was placed shallowly within the intercostal muscles along the pleura to perform thermal coagulation. Consequently, the impact on the pleura was limited, and as observed thoroscopically after this procedure, almost no pleural changes were noted. Of the cases reviewed, none required chest drain placement for postoperative pleural effusion in patients undergoing INTRA.

Table 2
Demographic, clinical and radiological features.

Number of cases	170 (Male: 139, Female:31)
Median age (Years)	17 (range; 12–66)
Average Haller index	5.17 ± 2.17
Average correction index	32.6 ± 14.4
Average body weight (Kg)	53.8 ± 10.5
Average MBI	18.9 ± 2.7
Average operation time (min.)	104.8 ± 33.3
Number of bars used	
One bar	3 cases
Two bars	92 cases
Three bars	75 cases

Long-term peripheral nerve recovery is another consideration. This heat (usually 70–90 °C in conventional RFA) causes coagulative necrosis and Wallerian degeneration of the axons, resulting in a temporary interruption of nerve conduction. However, the perineurium and epineurium are relatively preserved, allowing partial axonal regeneration over time. Excessive cautery temperatures and the length of the procedure may cause irreversible nerve damage; thus, appropriate temperature and duration settings are critical. The area of approximately 5 mm in diameter from the center of the needle is considered to undergo thermal degeneration at 70 °C and 60 s using a 20 G needle tip [19]. Our study found that 70 °C for 60 s provided effective pain relief while allowing peripheral nerve recovery over time.

Although ICA has demonstrated benefits in postoperative analgesia, its use is restricted in Japan due to insurance limitations. INTRA, which can be performed using readily available radiofrequency equipment, presents a viable alternative. Additionally, INTRA incurs lower equipment cost (116 US dollars for three needles) than ICA (2600 dollars for one probe). For intraoperative use of RFA, the surgeon must apply to the hospital's ethics committee; if approved, it can be used.

Another advantage is the ability to administer drugs such as steroids and local anesthetics through the needle tip near the intercostal costal nerves, which may improve postoperative pain control and nerve protection [15]. Although the occurrence of neuropathic pain has been reported after ICA [20], our results showed no case of neuropathic pain in long-term follow-up. It has been suggested that the incidence of neuropathic pain is higher in TRA than in ICA. However, a recent report showed that neuroma formation following radiofrequency ablation procedures is exceedingly rare [21]. Regarding heat complication on the blood vessels, the flow of blood prevents the temperature of the RF probe from rising when it comes into contact with a blood vessel.

Postoperative pain management is crucial not only in pectus excavatum repair but also in thoracic surgery, where approximately 57 % of patients experience chronic pain three months postoperatively [22]. The application of thoracoscopic intercostal nerve ablation techniques, such as INTRA, may provide long-term postoperative pain relief across various thoracic surgical procedures.

5. Conclusion

This study demonstrates that INTRA is a safe and effective method for postoperative pain control in pectus excavatum patients undergoing the Nuss procedure. As the first clinical application of INTRA in this setting, our primary objective was to evaluate its efficacy and safety. Although no direct comparison with previous cases was performed, future comparative studies with ICA are warranted to further evaluate its long-term benefits and broader clinical applications.

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